ECE 340 # Homework Assignment #2 **Fall 2026**

DUE: FRIDAY, SEPTEMBER 11, 2026

MAKE SURE YOUR **NAME** AND **NETID** ARE CLEARLY VISIBLE. SUBMIT ASSIGNMENTS AS A PDF (EITHER SCANNED OR TYPED). SHOW ALL WORK (PARTIAL CREDIT CANNOT BE GIVEN WITHOUT WORK). SHOW ALL UNITS. HOMEWORK MUST BE TURNED IN THROUGH GRADESCOPE. HOMEWORK SHOULD BE TURNED IN BEFORE 5:00PM ON THE DUE DATE. PER THE SYLLABUS LATE HOMEWORK IS NOT ACCEPTED.

1. CHARGE AND CARRIER ENERGY

Consider the band diagram below in Fig. 1.1:

(A) Which hole has the highest kinetic energy? Why?

(B) Which hole has the highest potential energy? Why?

(C) Which hole has the highest total energy? Why?

(D) Which of these valence band states has the highest probability to be occupied by an electron?

(E) Assuming the kinetic energy of the hole in region (3) is 2.5 eV, that the hole effective mass is $1.5\,m_0$, and that the electron effective mass in the conduction band is $0.75\,m_0$, what is the kinetic energy of an electron in the conduction band with the same momentum as this hole?

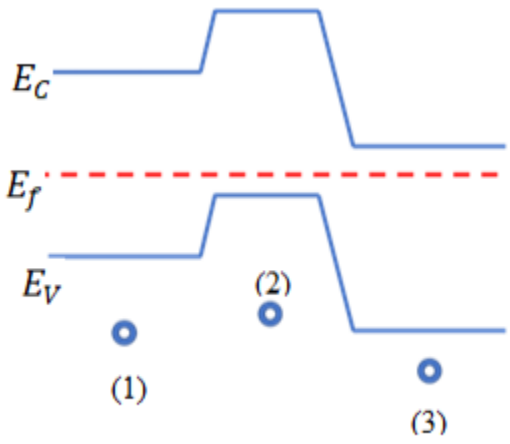


Figure 1.1: Energy Diagram for Q1

Solution **20 points in total, 4 points for each subproblem**

(A) Hole 2 has the highest KE as it is furthest away from the band edge. The hole can lose this KE as heat to relax to the band edge.

(B) Hole 3 has the highest PE since the valence band is furthest away from the Fermi level in that region.

(C) Hole 3 has the highest total energy. The sum of its KE and PE is the largest, and it is the furthest away from the Fermi level.

(D) According to the Fermi-Dirac distribution, state 3 has the highest probability of occupancy since it is the furthest from the Fermi level.

(E) We are given that the momenta are the same:

$$p_h = p_e \tag{1.1}$$

$$\sqrt{2(K.E.)_h \times m_h} = \sqrt{2(K.E.)_e \times m_e} \tag{1.2}$$

$$(K.E.)_e = (K.E.)_h \times \frac{m_h}{m_e} = 2.5\,eV \times \frac{1.5m_0}{0.75m_0} = 5\,eV \tag{1.3}$$

2. Materials and Band Structure

(A). Answer i and ii below:

i. Draw the $E-r$ (energy versus position) diagram for an insulator, a semiconductor, and a metal at 0 K. Label your diagram and show where the electrons reside.

ii. State the difference between the $E-r$ diagram of an insulator and a semiconductor at 0 K. How does the conduction band of a metal differ from those of an insulator and a semiconductor at 0 K?

(B). Consider the $E-k$ diagram of Fig. 2.1. The energy axis has units of eV. Answer (i)–(iv) based on this diagram.

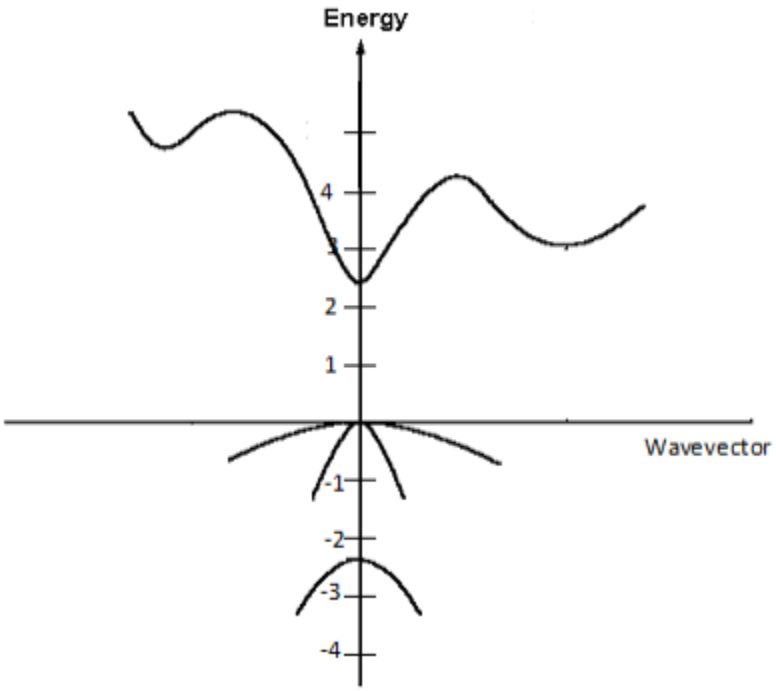


Figure 2.1: E–k diagram for Q2 (B–E)

i. Label each of the following bands on the diagram: conduction band (CB), light hole valence band (LHVB), heavy hole valence band (HHVB). State the reasons behind your labeling.

ii. Is this a direct or indirect bandgap material? Why? What is the value of the bandgap?

iii. An electron has a kinetic energy of 1.2 eV. Mark its position on the diagram.

iv. The curve near the conduction-band minima in the $E-k$ diagram follows the equation

$$E = 5 \times 10^{-38} k^2 \; Joules. \tag{2.1}$$

Find the electron effective mass.

Solution

20 points in total

(A) **6 points**

i. Answer is given in Fig. 2.2

ii. The bandgap of an insulator is larger than the bandgap of a semiconductor. At 0 K, metal has electrons in its conduction band whereas the conduction band of both insulator and semiconductor is empty.

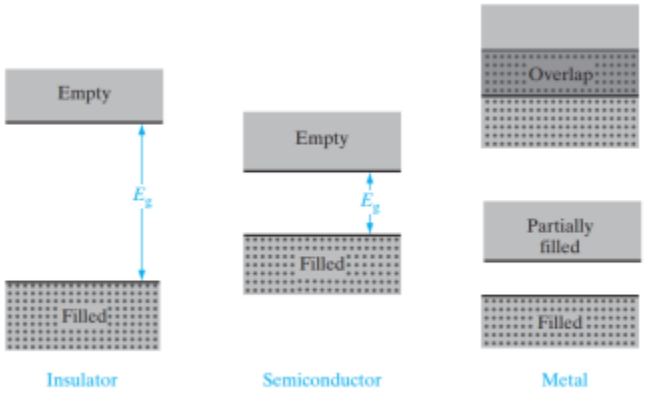


Figure 2.2: Answer for Q2(A)

(B) **14 points**

i. Please refer to Fig. 2.3. The conduction band curvature points downward near the minimum. It has a higher energy than the valence band. The valence band curvature points upward near the maxima. It has a lower energy than the valence band. As

$$m^* = \frac{\hbar^2}{\frac{d^2E}{dk^2}}, \tag{2.1}$$

the effective mass is inversely proportional to the curvature. The heavy hole band has less curvature than the light hole band.

ii. Direct bandgap. The conduction band minima (CVM) and valence band maxima (VBM) are at the same k value (on the y axis). The bandgap value is 2.5 eV.

iii. Please refer to Fig. 2.3. The correct marking should be positioned near 3.7 eV. If a horizontal line is drawn at 3.7 eV, any intersection points with the conduction band curve represent valid answers. Only one of these points is explicitly shown in the diagram.

iv. We are given:

$$E = \frac{\hbar^2 k^2}{2m^*} \tag{2.2}$$

$$\frac{d^2E}{dk^2} = \frac{\hbar^2}{m^*} \tag{2.3}$$

$$m^* = \frac{\hbar^2}{\frac{d^2E}{dk^2}} \tag{2.4}$$

From the given equation:

$$\frac{d^2E}{dk^2} = 2 \times 5 \times 10^{-38} \tag{2.5}$$

Using this:

$$m^* = \frac{\hbar^2}{2 \times 5 \times 10^{-38}} = 1.11 \times 10^{-31}\, kg = 0.12 m_0 \tag{2.6}$$

Here, $m_0 = 9.1 \times 10^{-31}\, kg$, the mass of an electron.

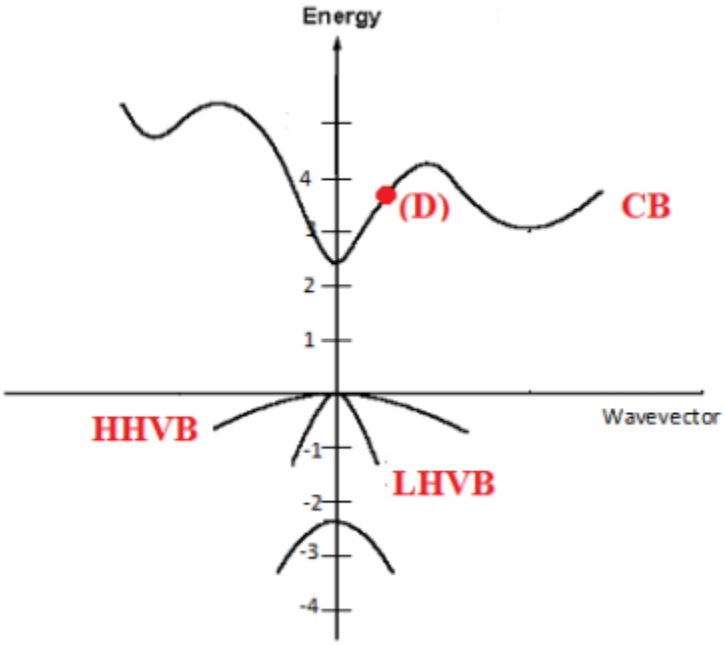


Figure 2.3: Answer for Q2(B,D)

3. Extrinsic and Intrinsic Materials

Silicon (Si) has an intrinsic carrier concentration $n_i = 1.5 \times 10^{10} cm^{-3}$ at 300 K.

(A). Assume the silicon sample is intrinsic at 300 K.

i. Find the electron concentration n.

ii. Find the hole concentration p.

iii. Briefly explain why $n = p$ in intrinsic material.

(B). Now the silicon is doped with acceptor atoms (B) at a concentration $N_A = 1.0 \times 10^{16} cm^{-3}$. Assume all acceptors are ionized, at thermal equilibrium, and no donor impurities.

i. Determine the majority carrier concentration.

ii. Calculate the minority carrier concentration using $np = ni^2$.

iii. Identify the majority and minority carriers.

(C). Explain with an energy band diagram and descriptive text why acceptor impurities can create free holes in the valence band at relatively low temperatures, while intrinsic carrier generation requires much higher temperatures.

Solution **20 points in total**

(A) **8 points**

i. $n = n_i = 1.5 \times 10^{10} cm^{-3}$.

ii. $p = n_i = 1.5 \times 10^{10} cm^{-3}$.

iii. In intrinsic material, every thermally generated electron in the conduction band corresponds to a hole left behind in the valence band. Therefore, $n = p = n_i$.

(B) **8 points**

i. $p \approx N_A$, $p = 1.0 \times 10^{16} cm^{-3}$.

ii. $np = n_i^2$, $n = \frac{n_i^2}{p} = \frac{(1.5\times10^{10})^2}{1.0\times10^{16}} = \frac{2.25\times10^{20}}{1.0\times10^{16}}$, $n = 2.25 \times 10^4 cm^{-3}$.

iii. Majority carriers: holes. Minority carriers: electrons.

(C) **4 points**

Acceptor impurities introduce energy levels slightly above the valence band. Because the energy separation is very small only a small amount of thermal energy is required to excite an electron from the valence band into the acceptor level, leaving a free hole behind in the valence band.

In contrast, intrinsic carrier generation requires electrons to cross the full band gap ($E_g \approx 1.12eV$ for Si), which requires much higher thermal energy. Therefore, acceptor-generated holes dominate conduction at room temperature.

4. Fermi–Dirac Statistics and Occupation Probability

(A). Calculate the probability of finding a filled electron state at the **conduction band edge** of an InP crystal ($E_g = 1.35\,eV$) at $T = 300\,\text{K}$ if the Fermi level (E_F) is at the following positions:

i. $E_F = E_C$

ii. $E_F = E_C - 6kT$

iii. $E_F = E_V$

(B). How does the probability of finding a filled electron state change (increase or decrease) as the Fermi level shifts from the conduction band towards the valence band in part (A)? Explain the reason behind this trend.

(C). Is the probability of finding an electron at the energy level $E_F + 1\,eV$ at 300 K higher or lower than at 100 K? Why?

Solution **20 points in total**

(A) **12 points**

We have to find the probability with $E = E_C$ for all three cases. The Fermi level changes for i, ii, and iii.

i. $E_F = E_C$. Since the Fermi level is at the conduction band, the probability of finding the electron there is $\frac{1}{2}$ by definition. (Or using the equation with $E = E_C$)

ii. $E_F = E_C - 6kT$

$$f(E) = \frac{1}{1+e^{\frac{E-E_F}{kT}}} = \frac{1}{1+e^{\frac{E_C-(E_C-6kT)}{kT}}} = \frac{1}{1+e^{6}} = 2.47\times10^{-3} \tag{4.1}$$

iii. $E_F = E_V$

$$f(E) = \frac{1}{1+e^{\frac{E-E_F}{kT}}} = \frac{1}{1+e^{\frac{E_C-E_V}{kT}}} = \frac{1}{1+e^{\frac{1.35eV}{0.0259eV}}} = 2.31\times10^{-23} \tag{4.2}$$

(B) **3 points**

The probability of finding a filled electron state decreases as the Fermi level moves from the conduction band towards the valence band. This results from the increasing gap between E (= E_C) and the Fermi level.

(C) **5 points**

This question can be answered either by using the Fermi–Dirac distribution mathematically or by considering the physical interpretation.

At higher temperatures, electrons possess greater thermal energy, which allows more of them to be excited to energy levels above the Fermi level. As the temperature increases, the Fermi–Dirac distribution becomes broader, meaning that the sharp transition around the Fermi energy is smeared out. This results in a higher probability of electrons occupying states with energies greater than E_F. In contrast, at lower temperatures, electrons have less thermal energy and tend to remain closer to the Fermi level, so the occupation of higher-energy states is significantly reduced. Therefore, increasing the temperature makes it more likely for electrons to be found at energies such as $E_F + 1eV$.